# Thermally induced deformation and conjugate heat transfer in a channel with a thermoelastic wall: formulation, reduced-order analysis and finite-element verification


**Abstract**

**Purpose** – To formulate, analyse and partly verify a conjugate heat-transfer problem in which a thin polymer strip, heated on its outer face, forms one wall of a laminar slit channel and bows because of through-wall thermal expansion, widening the passage.

**Design/methodology/approach** – A steady formulation with one consistent set of sign and interface conventions is given. Closed-form thermoelastic results (free curvature, bowing, buckling, beam–column amplification) define a baseline and admissibility guards. A finite-element method and a fixed-point coupling algorithm are specified; the wall is verified with two independent plane-strain codes, including a restrained, geometrically nonlinear case.

**Findings** – For water at $\mathrm{Re} = 50$ and a flux of $5\,\mathrm{kW/m^2}$ on a 3 mm PEEK-class strip, the strip bows away from the channel by $0.428\,H_0$ (reduced order) or $0.444\,H_0$ (finite element); the difference of 3.8 % vanishes for slender strips. Deformation and stress, not buckling, bound the flux. Indicative estimates suggest a roughly halved pressure drop and a higher wall excess temperature, while the return effect of the flow on the bow is small.

**Research limitations/implications** – No coupled fluid–structure result is reported; the flow-side estimates are reduced-order, the material data are representative; contact, heater and casing are idealised.

**Originality/value** – Thermally driven wall deformation in forced convection, a verified wall model and a specified, not yet executed, coupled solver.



**Keywords** Conjugate heat transfer, Thermoelasticity, Fluid–structure interaction, Finite element method, Flexible channel wall, Thermal bending

**Paper type** Research paper


## 1. Introduction

Compact heat exchangers, micro-reactors and electronics coolers are increasingly built with polymer walls, which combine low thermal conductivity with a high coefficient of thermal expansion. For the same imposed heat flux, such a wall sustains a through-thickness temperature difference that is far larger than that of a metallic wall of equal thickness, and linear thermoelasticity associates this difference with a free thermal curvature of the wall (Boley and Weiner, 1960, Timoshenko and Goodier, 1970). When the wall is as thick as, or thicker than, the channel is high, the curvature is not a small perturbation of the passage: it modifies the flow cross-section, and heat-transfer predictions that assume a fixed geometry become geometry-dependent in a thermally self-consistent way.

Fluid–structure interaction (FSI) with heat transfer has mostly been studied for deformation driven by the fluid. Soti et al. (2015) computed laminar channel flow past a flexible thin plate and showed that the flow-induced motion passively raises the heat transfer; Alsabery et al. (2019) and Saleh et al. (2025) treated transient natural convection in cavities with flexible walls; Christov et al. (2018) derived the steady relation between flow rate and pressure drop in a deformable shallow channel, again with pressure as the only load. The structure is thermally passive in these studies. Thermally driven structural motion has been

computed for heated flexible cantilevers in a differentially heated cavity (Kumar et al., 2026), a confined natural-convection configuration rather than a forced-convection channel. Conjugate heat transfer in rigid thick-wall channels, finally, has been studied (Rahimimotlagh et al., 2024), but there the wall does not move.

The configuration studied here differs in the mechanism and in the flow class. A finite segment of one wall of a laminar forced-convection slit is a thin thermoelastic strip, heated on its outer face. Conduction across the strip sets up a temperature difference; the associated eigen-curvature, acting on a strip with pinned ends, bows it *away* from the passage; the widened passage changes the near-wall velocity and temperature gradients and therefore the interfacial heat flux, which is one of the boundary data of the conduction problem in the strip. Within the literature examined for this work we did not find a study that combines (i) the thermal eigen-curvature of the wall as the principal deformation mechanism, (ii) a closed steady loop through the conjugate interface, and (iii) a confined forced-convection channel. This positioning is deliberately bounded: the literature search was not exhaustive, and it is not a claim of priority.

The paper makes five contributions, and states clearly what it does not do.

1. A complete steady formulation with one consistent set of sign and interface conventions (Section 2).
2. A reduced-order thermoelastic analysis of the strip in closed form: free curvature, pinned bowing, thermal membrane force and buckling, the beam–column amplification of the bow, and the ratio of thermal to hydraulic deflection (Section 3).
3. A baseline operating point evaluated from first principles, with the heat-flux ceilings imposed by deformation, stress and buckling, and an *indicative* estimate of what the deformation does to the flow (Section 4).
4. A specified finite-element formulation and fixed-point algorithm for the coupled problem, with the treatment of the interface variables made explicit (Section 5.1).
5. A plane-strain Q8 verification of the wall idealisation: a mesh-sensitivity study of the free bow at a fixed sampling height, and an axially restrained, geometrically nonlinear solution of the baseline wall (Section 5.4).

**No coupled fluid–structure computation is reported.** The coupled solver is specified, not executed; every statement about the flow in Section 4 is a reduced-order estimate and is labelled as such. The paper is organised accordingly: formulation, analysis, baseline, finite element method and verification, and limitations. Symbols are listed in Table IX.

## 2. Problem formulation

### *2.1 Configuration and assumptions*

The configuration is two-dimensional (Figure 1). The fluid reference domain is $\Omega_f^0 = \{-L_e \le X \le L+L_o,\ 0 \le Y \le H_0\}$: a rigid entrance section, a section of length $L$ facing the deformable wall, and a rigid exit section. The only deformable body is the solid strip $\Omega_s^0 = [0,L] \times [H_0, H_0+d]$, of thickness $d$ and unit out-of-plane depth. Its lower face $Y = H_0$ is the conjugate interface $\Gamma_{\mathrm{if}}$; its upper face $Y = H_0 + d$ is the outer face $\Gamma_h$, on which the thermal load is imposed. The strip ends $X = 0$ and $X = L$ are pinned. Rigid, adiabatic casing sections carry the strip at its ends, and the lower channel wall is rigid and adiabatic. The geometry and the operating point are given in Table I.

The following assumptions define the model; each is checked for the baseline in Section 4.

**A1** *Plane strain.* The out-of-plane dimension is large compared with $H_0$ and $d$ and the loading is uniform along it, so the fields are two-dimensional and $\varepsilon_{zz} = 0$ in the solid (Boley and Weiner, 1960, Timoshenko and Goodier, 1970).

**A2** *Newtonian, constant-property water* at near-ambient temperature. Laminar, steady flow ($\mathrm{Re}_{D_h} = 50$);

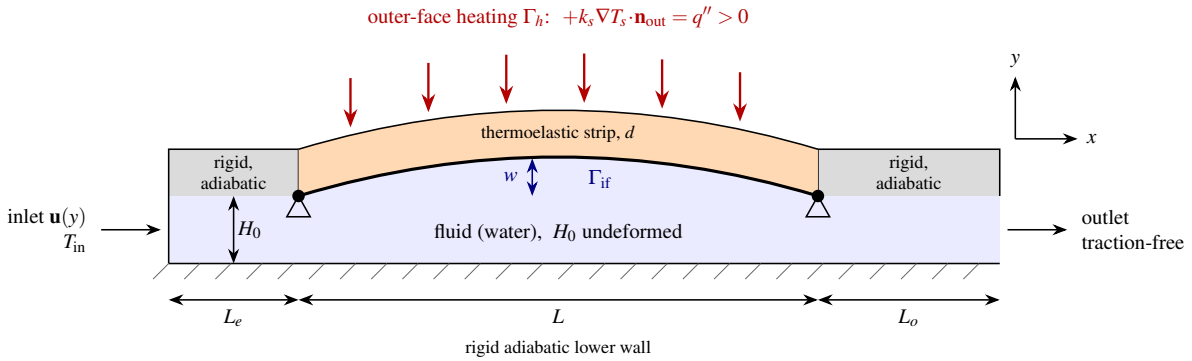



**Figure 1.** Configuration (schematic, not to scale; the strip thickness and the bow are exaggerated). A thermoelastic strip of length $L$ and thickness $d$ forms part of the upper wall of a slit channel of undeformed height $H_0$. The outer face $\Gamma_h$ receives a prescribed heat flux $q'' > 0$ into the strip. With pinned ends the strip bows away from the channel: the interface deflection $w$ is positive away from the channel and the passage widens.

**Table I.** Geometry and operating point of the baseline (design values).

| Quantity | Symbol | Value | Note |
|---|---|---|---|
| Undeformed channel height | $H_0$ | 0.50 mm | design input |
| Hydraulic diameter (slit) | $D_h = 2H_0$ | 1.0 mm | derived |
| Deformable (heated) length | $L$ | 30 mm | design |
| Entrance / exit length | $L_e$, $L_o$ | 10 mm, 20 mm | design |
| Strip thickness | $d$ | 3.0 mm | design |
| Mean inlet velocity | $U$ | 0.050 m/s | design |
| Imposed outer-face heat flux | $q''$ | 5.0 kW/m$^2$ | design |

viscous dissipation and buoyancy are negligible at the baseline (Section 4).

**A3** *Homogeneous, isotropic, linearly thermoelastic solid* of constant $E$, $\nu$, $\alpha$, $k_s$. The values are representative of an unfilled PEEK-class polymer (Table III); they are not frozen against a specific datasheet, and every derived number scales with them.

**A4** *Small strains and moderate rotations*, with the wall equation carried to the von Kármán beam–column level of Section 3.4.

**A5** *Steady flow and energy, quasi-static solid.* The structural wave-crossing time of the strip, $d\sqrt{\rho_s/E} \approx 1.8\,\mu$s, is many orders of magnitude shorter than the transport times, so inertia is omitted.

**A6** *Perfect thermal contact* at $\Gamma_{\text{if}}$: temperature and normal heat flux are continuous; no contact resistance.

**A7** *Ideal pinned ends.* At $X = 0, L$ the vertical displacement vanishes, the mean axial displacement over the end edge vanishes, the applied moment is zero, and the heat flux is zero. This is a pin and not a clamp: a clamp forbids end rotation, and a clamped strip at uniform curvature does not bow at all (Section 3.1).

**A8** *Idealised heater.* The heater is represented only by a uniform flux $q''$ entering the solid through $\Gamma_h$. Heater construction, heater–wall contact resistance and heat loss from the outer face are not modelled; all of $q''$ is assumed to enter the strip.

**A9** *Rigid casing as a support, not as a contact.* The casing fixes the end conditions and the reference pressure of the cavity above the strip (equal to the outlet datum). The model contains no contact between the deformed strip and the casing and no casing clearance; the cavity above the strip is assumed to leave room for the computed deflection.

### 2.2 Governing equations

*Fluid.* On the deformed fluid domain $\Omega_f$ (Section 2.3), steady incompressible flow with constant properties obeys, in the standard form (Bird et al., 1960),

$$\nabla\cdot\boldsymbol{u} = 0, \tag{1}$$

$$\rho\,(\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = -\nabla p + \mu\nabla^2\boldsymbol{u}, \tag{2}$$

$$\rho c_p\,(\boldsymbol{u}\cdot\nabla)T_f = k_f\nabla^2 T_f. \tag{3}$$

*Solid.* On the reference configuration $\Omega_s^0$ steady conduction (Carslaw and Jaeger, 1947) and linear thermoelastic equilibrium without inertia hold,

$$\nabla\cdot(k_s\nabla T_s) = 0, \tag{4}$$

$$\nabla\cdot\boldsymbol{\sigma} = \mathbf{0}, \qquad \boldsymbol{\sigma} = 2\mu_L\boldsymbol{\varepsilon} + \lambda(\operatorname{tr}\boldsymbol{\varepsilon})\boldsymbol{I} - \beta\theta\boldsymbol{I}, \qquad \boldsymbol{\varepsilon} = \tfrac{1}{2}\left(\nabla\boldsymbol{d} + \nabla\boldsymbol{d}^{\mathsf{T}}\right), \tag{5}$$

with $\theta = T_s - T_{\mathrm{ref}}$, Lamé parameters $\lambda, \mu_L$ and thermoelastic pressure coefficient $\beta = E\alpha/(1-2\nu)$ (Boley and Weiner, 1960). In plane strain the out-of-plane stress is $\sigma_{zz} = \nu(\sigma_{xx}+\sigma_{yy}) - E\alpha\theta$ and is part of the stress state; all stresses reported below are total Cauchy stresses. The deformation-heating source of the transient thermoelastic energy equation vanishes identically in a steady state, so (4) is exact for the steady problem.

*Reduced wall equation.* For design estimates the strip is treated as an Euler–Bernoulli strip of flexural rigidity per unit depth $D_s = Ed^3/[12(1-\nu^2)]$, the wide-strip value appropriate to plane strain,

$$D_s\,w'''' = p(x), \qquad w(0) = w(L) = w''(0) = w''(L) = 0, \tag{6}$$

where $p$ is the fluid pressure acting on the strip (positive pushes the strip away from the channel). This equation is used for estimates and benchmarks only; the two-dimensional system (4)–(5) is the model.

### 2.3 Deformed domain

The fluid problem is posed on the domain occupied by the fluid in the converged steady state. Given the solid displacement on $\Gamma_{\mathrm{if}}^0$, a mesh displacement $\widehat{\boldsymbol{w}}$ is obtained on $\Omega_f^0$ from a pseudo-elastic extension problem with $\widehat{\boldsymbol{w}} = \boldsymbol{d}$ on $\Gamma_{\mathrm{if}}^0$ and $\widehat{\boldsymbol{w}} = \mathbf{0}$ on rigid boundaries, and $\Omega_f = \{X + \widehat{\boldsymbol{w}}(X)\}$. No time derivative appears, so no mesh velocity or geometric-conservation-law constraint arises in the steady problem (Lesoinne and Farhat, 1996).

### 2.4 Sign conventions, interface and boundary conditions

A single unit normal $\boldsymbol{n}$, pointing from the fluid into the solid, is used on $\Gamma_{\mathrm{if}}$. Pressure is positive in compression and stress is positive in tension. The coordinate $y$ points from the fluid into the solid. The deflection $w$ is measured *away from the channel*, $w = +d_2$ with $d_2$ the displacement in $+y$; the thermal curvature $\kappa_{\mathrm{th}} > 0$ is the curvature that bows the strip away from the channel, and within small slopes $\kappa = -w''$. The heat flux vector is $\boldsymbol{q} = -k\nabla T$, and $q'' > 0$ denotes heat *supplied* to the solid.

On the deformed interface the subsystems exchange kinematics, traction and heat:

$$\boldsymbol{u} = \mathbf{0} \qquad \text{(no slip, steady form of } \boldsymbol{u} = \partial_t\boldsymbol{d}\text{)}, \tag{7}$$

$$\boldsymbol{\sigma}\boldsymbol{n} = (-p\boldsymbol{I} + 2\mu\boldsymbol{D}(\boldsymbol{u}))\,\boldsymbol{n} \qquad \text{(traction continuity)}, \tag{8}$$

$$T_f = T_s \qquad \text{(temperature continuity)}, \tag{9}$$

$$-k_f\nabla T_f\cdot\boldsymbol{n} = -k_s\nabla T_s\cdot\boldsymbol{n} \qquad \text{(normal heat-flux continuity)}, \tag{10}$$

with $\boldsymbol{D}(\boldsymbol{u}) = \frac{1}{2}(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^{\mathsf{T}})$. The exterior conditions are listed in Table II. Condition B5 makes the sign of the thermal load explicit: with the outward normal $\boldsymbol{n}_{\mathrm{out}} = +\boldsymbol{e}_y$ of the solid on $\Gamma_h$, the flux $\boldsymbol{q}\cdot\boldsymbol{n}_{\mathrm{out}} = -k_s\partial_y T_s$

**Table II.** Boundary conditions. $\boldsymbol{n}_{\text{out}}$ is the outward normal of the solid.

| ID | Boundary | Condition |
|---|---|---|
| B1 | inlet $x = -L_e$ | $\boldsymbol{u} = 6U\frac{y}{H_0}\left(1-\frac{y}{H_0}\right)\boldsymbol{e}_x$, $T_f = T_{\text{in}}$ |
| B2 | outlet $x = L + L_o$ | traction-free (do-nothing) outflow, $\partial_n T_f = 0$ |
| B3 | lower wall $y = 0$ | $\boldsymbol{u} = \boldsymbol{0}$, $\partial_n T_f = 0$ (rigid, adiabatic) |
| B4 | rigid casing sections | $\boldsymbol{u} = \boldsymbol{0}$, $\partial_n T_f = 0$ |
| B5 | outer face $\Gamma_h^0$ | $+k_s \nabla T_s \cdot \boldsymbol{n}_{\text{out}} = q''$ with $q'' > 0$ heat supplied to the solid |
| B6 | strip ends $X = 0, L$ | $d_2 = 0$, $\langle d_1 \rangle = 0$ (edge mean), zero applied moment, $\partial_n T_s = 0$ |
| B7 | remaining solid faces | zero traction, zero heat flux |
| B8 | interface $\Gamma_{\text{if}}$ | (7)–(10) |

**Table III.** Material properties used in all calculations.

| Property | Symbol | Value | Unit |
|---|---|---|---|
| Water density, viscosity | $\rho$, $\mu$ | $1.0\times10^3$, $1.0\times10^{-3}$ | kg/m$^3$, Pa s |
| Water specific heat, conductivity | $c_p$, $k_f$ | $4.18\times10^3$, 0.60 | J/(kg K), W/(m K) |
| Solid Young's modulus, Poisson ratio | $E$, $\nu$ | $3.6\times10^9$, 0.40 | Pa, – |
| Solid expansion coefficient | $\alpha$ | $5.0\times10^{-5}$ | 1/K |
| Solid conductivity | $k_s$ | 0.25 | W/(m K) |
| Solid yield stress (guard only) | $\sigma_y$ | $9\times10^7$ | Pa |

equals $-q''$ (inward), so $k_s\,\partial_y T_s = +q'' > 0$: the temperature rises towards the outer face, which is therefore the hot face.

### *2.5 Material data*

Table III lists the properties. Water values are standard near-ambient data. The solid values are representative of unfilled PEEK and are datasheet-representative, not grade-specific.

## 3. Reduced-order thermoelastic analysis of the strip

This section derives the closed-form relations used to size the deformation before any coupled computation. All of them follow from the model of Section 2 under the stated assumptions; the numerical values at the baseline are collected in Section 4.

### *3.1 Free thermal curvature and the sign chain*

Let $\eta$ be the coordinate across the strip measured from the mid-plane (positive towards the outer face), and let the temperature be linear, $\theta(\eta) = \bar{\theta} + \Gamma\eta$, with $\Gamma = \Delta T_t/d > 0$ by condition B5. For a thin strip with traction-free faces $\sigma_{yy} \simeq 0$, and plane strain gives $\sigma_{xx} = E'[\varepsilon_{xx} - (1+\nu)\alpha\theta]$ with $E' = E/(1-\nu^2)$. With $\varepsilon_{xx} = \varepsilon_0 + \kappa\eta$, a vanishing axial force resultant gives $\varepsilon_0 = (1+\nu)\alpha\bar{\theta}$, and a vanishing moment resultant gives the free thermal eigen-curvature

$$\kappa_{\text{th}} = (1+\nu)\,\alpha\,\frac{\Delta T_t}{d} = (1+\nu)\,\alpha\,\frac{q''}{k_s}, \qquad \Delta T_t = \frac{q''d}{k_s}, \tag{11}$$

where the second form uses one-dimensional steady conduction across the strip (Carslaw and Jaeger, 1947) and is the plane-strain form of the classical result (Boley and Weiner, 1960). Two consequences follow. First, $\kappa_{\text{th}}$ does not depend on $d$: the through-thickness gradient adjusts to cancel the thickness. Second, a strip clamped at both ends carries the constant restoring moment $D_s\kappa_{\text{th}}$ and does not bow ($w \equiv 0$); the bowing mechanism exists only for pinned ends or for an axially non-uniform $\kappa_{\text{th}}(x)$, which

fixes the support choice A7.

The sign chain is, in order: (i) $q'' > 0$ supplies heat to the outer face (B5); (ii) the temperature therefore rises towards the outer face ($\Gamma > 0$), which is the hot face; (iii) the hot fibres elongate more, so the strip curves with the outer face convex, $\kappa_{\mathrm{th}} > 0$; (iv) with pinned ends this is an arch whose mid-span moves *away* from the channel, $w > 0$; (v) the passage height $H = H_0 + w$ increases. The same chain read backwards shows that a bow *into* the passage would require heating the fluid-facing face, which is a different model (Section 6).

### *3.2 Pinned bowing at negligible axial force*

With $w'' = -\kappa_{\mathrm{th}}$ and $w(0) = w(L) = 0$,

$$w(x) = \frac{\kappa_{\mathrm{th}}}{2} x(L-x), \qquad w_{\max}^{\mathrm{lin}} = \frac{\kappa_{\mathrm{th}} L^2}{8}. \tag{12}$$

This is exact within thin-strip theory when the axial force is negligible, which is the state of a strip free to expand axially.

### *3.3 Thermal membrane force and buckling*

With the axial displacement blocked at both ends the mean axial strain vanishes, and integration across the thickness gives the compressive membrane force per unit depth

$$N_\theta = -\frac{E\, d\, \alpha \bar{\theta}}{1-\nu}, \qquad N_c = |N_\theta|. \tag{13}$$

The pinned–pinned Euler load of the wide strip is $N_{\mathrm{cr}} = \pi^2 D_s / L^2$ (Timoshenko and Gere, 1961). The wide-strip rigidity is the consistent one because plane strain suppresses the anticlastic curvature mode. Setting $N_c = N_{\mathrm{cr}}$ gives the critical mean temperature rise

$$\bar{T}_{\mathrm{cr}} = \frac{\pi^2 d^2}{12\,\alpha(1+\nu)L^2}, \qquad M_b = \frac{\bar{T}_{\mathrm{cr}}}{\bar{\theta}}, \tag{14}$$

where the buckling margin $M_b$ is used as a guard. The mean wall rise is estimated as $\bar{\theta} = \Delta T_t/2 + q''/h + \Delta T_b/2$: half the through-wall drop, the interface excess over the local bulk temperature with the rigid-slit coefficient $h = \mathrm{Nu}_{H1} k_f / D_h$ for one heated wall (Shah and London, 1978), and half the bulk rise $\Delta T_b = q'' L/(\rho U H_0 c_p)$.

### *3.4 Beam–column bowing at finite membrane force*

A pinned strip carrying the compressive force $N_c$ couples its deflection to that force at leading order. Retaining the geometric stiffness of the membrane force and freezing the force at its blocked-state value, the transverse equilibrium of the strip, with total moment $M = D_s(w'' + \kappa_{\mathrm{th}})$ vanishing at the pins, is

$$D_s\, w'''' + N_c\, w'' = 0, \qquad w(0) = w(L) = 0, \quad w''(0) = w''(L) = -\kappa_{\mathrm{th}}. \tag{15}$$

With $\mu^2 = N_c/D_s$ the solution is

$$w(x) = \frac{\kappa_{\mathrm{th}}}{\mu^2}\left[\frac{1-\cos\mu L}{\sin\mu L}\sin\mu x + \cos\mu x - 1\right], \qquad w_{\max} = \frac{\kappa_{\mathrm{th}}}{\mu^2}\left[\sec\frac{\mu L}{2} - 1\right], \tag{16}$$

which reduces to (12) as $\mu L \to 0$ and diverges at $\mu L = \pi$, the Euler threshold (14): the bowing law and the buckling limit are two limits of one relation. The force is not in fact frozen. With both ends blocked, the mean axial strain contains the arc-stretching term $\frac{1}{2}\langle w'^2\rangle$, so the signed force acquires the correction

$$N_\theta(w) = N_\theta + \frac{Ed}{2(1-\nu^2)}\langle w'^2\rangle, \qquad N_c^{\mathrm{sc}} = |N_\theta| - \frac{Ed}{2(1-\nu^2)}\langle w'^2\rangle, \tag{17}$$

which relaxes the compression. Because the family (16) holds for every fixed force, (17) is a scalar fixed-point problem for $N_c^{\mathrm{sc}}$.

Two deflections are therefore defined, and they answer different questions. The *frozen-force* value $w_{\max}$ of (16) with $N_c = |N_\theta|$ is the exact solution of the stated equation (15); because it neglects the relaxation it overestimates the deflection and is a conservative upper bound. The *self-consistent* value solves (16) with $N_c^{\mathrm{sc}}$ and is the von Kármán equilibrium of the reduced strip. It is also what the coupled algorithm of Section 5.2 converges to, since it re-evaluates the membrane force at every iteration. The self-consistent value is therefore quoted as the model's best estimate, and the frozen-force value as the upper bound; both are reported. The two differ by 2.3 % at the baseline, 3.0 % and 3.9 % at the heat fluxes of guards G1 and G2 (Section 3.6), and 17 % at $N_c/N_{\mathrm{cr}} = 0.5$, where the frozen-force form should no longer be used.

### *3.5 Hydraulic loading and the thermal-to-hydraulic ratio*

Integrating plane Poiseuille flow (Bird et al., 1960, Shah and London, 1978) gives the slit pressure drop

$$\Delta p = \frac{12\mu U L}{H_0^2}. \tag{18}$$

With the outlet datum $p = 0$ and the casing cavity at the same reference pressure (A9), the gauge pressure on the strip falls linearly from $p(0) = p(L) + \Delta p$ at its upstream end to $p(L) = 12\mu U L_o/H_0^2$ at its downstream end, with mean $\bar{p} = \frac{1}{2}[p(0) + p(L)]$. For pinned ends, a uniform load $\bar{p}$ acting in the same sense as the thermal bow deflects the strip by $w_p = 5\bar{p}L^4/(384 D_s)$ at the linear level (Timoshenko and Goodier, 1970); the mild variation of the load is neglected in this estimate. The ratio of the thermal to the hydraulic deflection of the *same* strip is

$$\Gamma_p = \frac{w_{\mathrm{th}}}{w_p}, \qquad \Gamma_p^{\mathrm{lin}} = \frac{w_{\max}^{\mathrm{lin}}}{w_p} = \frac{48\,\kappa_{\mathrm{th}} D_s}{5\,\bar{p} L^2}, \tag{19}$$

where $\Gamma_p^{\mathrm{lin}}$ uses both deflections at zero membrane force. A second like-for-like form, $\Gamma_p^{N_c}$, computes both deflections with the same compressive force $N_c$: the thermal deflection from (16) and the pressure deflection from the beam–column equation $D_s w'''' + N_c w'' = \bar{p}$. Both forms compare two responses of one structure in one state and are therefore directly comparable. A ratio that places the thermal deflection *with* the beam–column amplification over a pressure deflection *without* it mixes two states of the structure and is not used. Because $\kappa_{\mathrm{th}}$ is independent of $d$ while $D_s \propto d^3$ and $\bar{p} \propto U H_0^{-2}$, $\Gamma_p$ grows with wall thickness and falls with velocity and with decreasing channel height.

### *3.6 Admissibility guards*

Five guards bound the regime in which the reduced-order model and the weak form of Section 5.1 are used. (G1) $w_{\max}/H_0 \le 0.5$, so that the deformed fluid mesh keeps positive Jacobians. (G2) $\sigma_{\max} \le 0.2\,\sigma_y$, with $\sigma_{\max}$ the sum of the bending stress $6|M_{\mathrm{mid}}|/d^2$ at mid-span and the membrane stress $E\alpha\bar{\theta}/(1-\nu)$. (G3) $M_b \ge 2$. (G4) $\Gamma_p \ge 5$, the threshold above which the thermal bow is the dominant deformation. (G5) $\mathrm{Re}_{D_h} \le 100$, $\mathrm{Br} < 10^{-2}$ and a Richardson number small enough for the neglect of buoyancy, with $\mathrm{Ri} = g\beta_f \Delta T^* H_0/U^2$ and $\Delta T^* = q'' H_0/k_f$.

## 4. Baseline results of the reduced-order model

All values in this section are computed by the program `code/hff_model.py` from the inputs of Tables I and III and the relations of Section 3; Table IV collects them. Quantities that depend on the solid properties scale with the datasheet values, which are not frozen (A3).

### *4.1 Deformation of the passage*

The strip bows away from the channel by $w_{\max} = 2.14 \times 10^{-4}$ m, which is $0.428\,H_0$, in the self-consistent state; the frozen-force upper bound is $0.438\,H_0$ (Figure 2a). The passage height at mid-span is $H =$

**Table IV.** Reduced-order baseline at the design point of Table I. The deflection and passage height are given for the self-consistent state (sc) and, as an upper bound, for the frozen-force state (ff).

| Quantity | Expression | Value |
|---|---|---|
| Reynolds, Péclet numbers | $2\rho U H_0/\mu$; Re Pr | 50; 348 (Pr = 6.97) |
| Rigid-slit pressure drop | $12\mu UL/H_0^2$ (18) | 72 Pa; wall shear 0.6 Pa |
| Through-wall temperature drop | $\Delta T_t = q''d/k_s$ | 60 K |
| Bulk rise; interface excess | $\Delta T_b$; $q''/h$ with $h = 3231\,\mathrm{W/(m^2\,K)}$ | 1.44 K; 1.55 K |
| Mean wall temperature rise | $\bar{\theta} = \Delta T_t/2 + q''/h + \Delta T_b/2$ | 32.3 K |
| Free thermal curvature | $(1+\nu)\alpha q''/k_s$ (11) | 1.4/m |
| Linear bow | $\kappa_{\mathrm{th}} L^2/8$ (12) | $1.57\times10^{-4}$ m $= 0.315\,H_0$ |
| Membrane force; Euler load | $N_c$ (13); $N_{\mathrm{cr}} = \pi^2 D_s/L^2$ | $2.90\times10^4$; $1.06\times10^5$ N m$^{-1}$; $N_c/N_{\mathrm{cr}} = 0.275$ |
| Critical rise; buckling margin | $\bar{T}_{\mathrm{cr}}$; $M_b$ (14) | 117 K; 3.64 |
| Beam–column parameter | $\mu L = L\sqrt{N_c/D_s}$ | 1.65 |
| Bow, frozen force (ff) | (16) | $2.19\times10^{-4}$ m $= 0.438\,H_0$ |
| Bow, self-consistent (sc) | (16), (17) | $2.14\times10^{-4}$ m $= 0.428\,H_0$ |
| Mid-span passage height $H_0 + w_{\max}$ | derived | $1.428\,H_0$ (sc); $1.4377\,H_0$ = 0.719 mm (ff) |
| Slope, strains (ff) | $\max\lvert w'\rvert$; $\frac{1}{2}\langle w'^2\rangle$; bending fibre strain | 0.028; $1.39\times10^{-4}$; $9.89\times10^{-4}$ |
| Stresses (ff) | bending; membrane; sum | 4.2 MPa; 9.7 MPa; 13.9 MPa = $0.15\,\sigma_y$ |
| Gauge pressure on the strip | $p(0)$; $p(L)$; $\bar{p}$ | 120 Pa; 48 Pa; 84 Pa |
| Hydraulic deflection | $5\bar{p}L^4/(384 D_s)$ | $9.19\times10^{-8}$ m (zero force); $1.27\times10^{-7}$ m (with $N_c$) |
| Thermal-to-hydraulic ratio | $\Gamma_p^{\mathrm{lin}}$; $\Gamma_p^{N_c}$ (19) | 1714; 1726 |
| Buoyancy, dissipation | Ri; Br | $1.7\times10^{-3}$; $1.0\times10^{-6}$ |

$H_0 + w_{\max} = 1.428\,H_0$ (sc) or $1.4377\,H_0$ (ff), that is 0.719 mm against the undeformed 0.50 mm. This is a *derived beam-model passage height*, not a prescribed clearance: it is the sum of the design height $H_0$ and the computed deflection, and nothing in the model constrains it from above (assumption A9). A two-dimensional finite-element solution of the restrained wall (Section 5.4.2) gives $0.444\,H_0$, 3.8 % above the self-consistent value. The beam–column amplification over the linear bow is 1.39, consistent at leading order with the Euler factor $1/(1-N_c/N_{\mathrm{cr}}) = 1.38$. The strains and the slope are small ($\max\lvert w'\rvert = 0.028$, bending fibre strain $9.89\times10^{-4}$), which supports the small-strain, moderate-rotation level of A4.

### *4.2 Guards and heat-flux ceilings*

The baseline lies inside all five guards of Section 3.6: $w_{\max}/H_0 = 0.438$ (ff) against 0.5; $\sigma_{\max} = 0.15\,\sigma_y$ against 0.2; $M_b = 3.64$ against 2; $\Gamma_p = 1714$ against 5; $\mathrm{Re}_{D_h} = 50$, $\mathrm{Br} = 1.0\times10^{-6}$ and $\mathrm{Ri} = 1.7\times10^{-3}$. All but the last are exceeded in the order deformation, stress, buckling as $q''$ increases, because $\kappa_{\mathrm{th}}$ and $\bar{\theta}$ both scale with $q''$. In the frozen-force state G1 is reached at $q'' = 5.5\,\mathrm{kW/m^2}$ ($N_c/N_{\mathrm{cr}} = 0.30$), G2 at 6.0 kW/m$^2$ ($w_{\max}/H_0 = 0.56$), and the buckling margin $M_b = 2$ at 9.1 kW/m$^2$, where $w_{\max}/H_0 = 1.16$ would long since be unacceptable. The binding constraints are therefore deformation and stress, with buckling the outer ceiling, and the baseline flux is already 9 % below the first of them.

Buoyancy and viscous dissipation do not enter the baseline: $\mathrm{Ri} = 1.7\times10^{-3}$ and $\mathrm{Br} = 1.0\times10^{-6}$. Stratification can destabilise horizontal flows when it is strong, as studied for a fluid with large property variations by Draskic et al. (2025); that regime is not approached here.

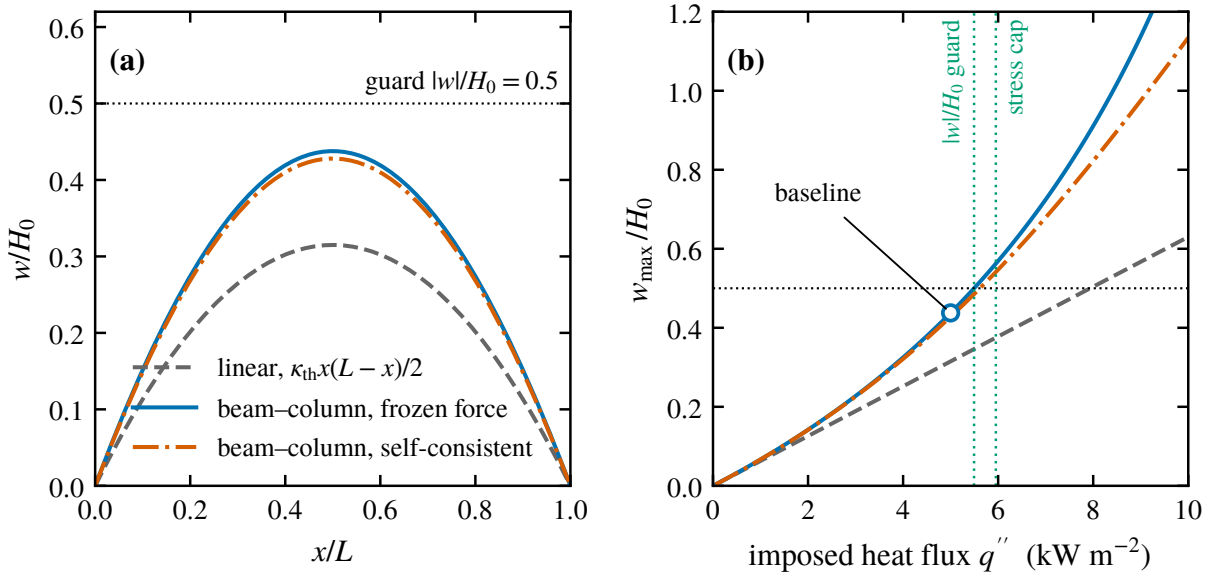



**Figure 2.** Bowing of the strip (positive away from the channel). (a) Deflection profile at the baseline: linear estimate (12), frozen-force beam–column (16) and self-consistent beam–column (16)–(17). (b) Mid-span deflection against the imposed flux; vertical dotted lines mark the fluxes at which guards G1 ($|w|/H_0 = 0.5$) and G2 (stress cap) are reached. The marker is the baseline.

The stress is dominated by the compressive membrane component, 9.7 MPa, a by-product of the axial constraint that also produces the buckling limit; the bending component is 4.2 MPa.

### *4.3 Role of the wall material*

Because $\kappa_{th} = (1+\nu)\alpha q''/k_s$, the choice of wall material sets the size of the effect. For typical austenitic stainless steel ($k_s \approx 16\,\mathrm{W/(m\,K)}$, $\alpha \approx 1.7\times10^{-5}/\mathrm{K}$, $\nu = 0.30$) at the same flux and geometry, $\kappa_{th} = 6.9\times10^{-3}$ 1/m and $w^{lin}_{max}/H_0 = 1.6\times10^{-3}$: about 203 times smaller than for the polymer, a physically real deformation but irrelevant to the convective field. The combination of low conductivity and high expansion coefficient of the polymer class, not any fluid-side property, selects the design space.

### *4.4 Indicative consequences for the flow*

The deformation is large enough to change the passage substantially, so two flow-side consequences are estimated. Both are reduced-order and indicative only; the coupled computation is required to quantify them. *Hydraulic.* With the flow rate per unit depth $UH_0$ fixed and the local height $H(x) = H_0 + w(x)$, the lubrication estimate is $\Delta p = 12\mu U H_0 \int_0^L \mathrm{d}x/H^3$, which is 36 Pa, a factor 0.50 of the rigid-slit value (18), a drop of about 50 per cent (Figure 3b). The lubrication parameter $\mathrm{Re}_{H_0} H_0/L = 0.42$ is not small, so the value indicates that the hydraulic effect is of order one rather than giving its magnitude. *Thermal.* With a quasi-developed local coefficient $h(x) = \mathrm{Nu}_{H1} k_f/[2H(x)]$ the mean interface excess $q''/h$ rises from 1.55 K to 1.99 K (a factor 1.29). The thermal entry length is about 17.4 mm, so $L/L_T = 1.7$ and the quasi-developed assumption is not justified; this too is an order-of-magnitude indicator.

*Feedback onto the deflection.* The larger interface excess raises $\bar{\theta}$ by about 0.45 K out of 32.3 K, which increases $w_{max}$ by only 0.5 %; the hydraulic load changes the deflection by 0.06 % of the thermal bow ($1/\Gamma_p$). At the baseline the effect of the wall on the flow is therefore large, whereas the effect of the flow back on the wall is small, because $\kappa_{th}$ depends on $q''$ and on the wall properties only (equation (11)) and the thermal loop enters through $\bar{\theta}$ alone. Whether the closed loop matters elsewhere in the parameter space is a question for the coupled computation, not for this reduced-order model.

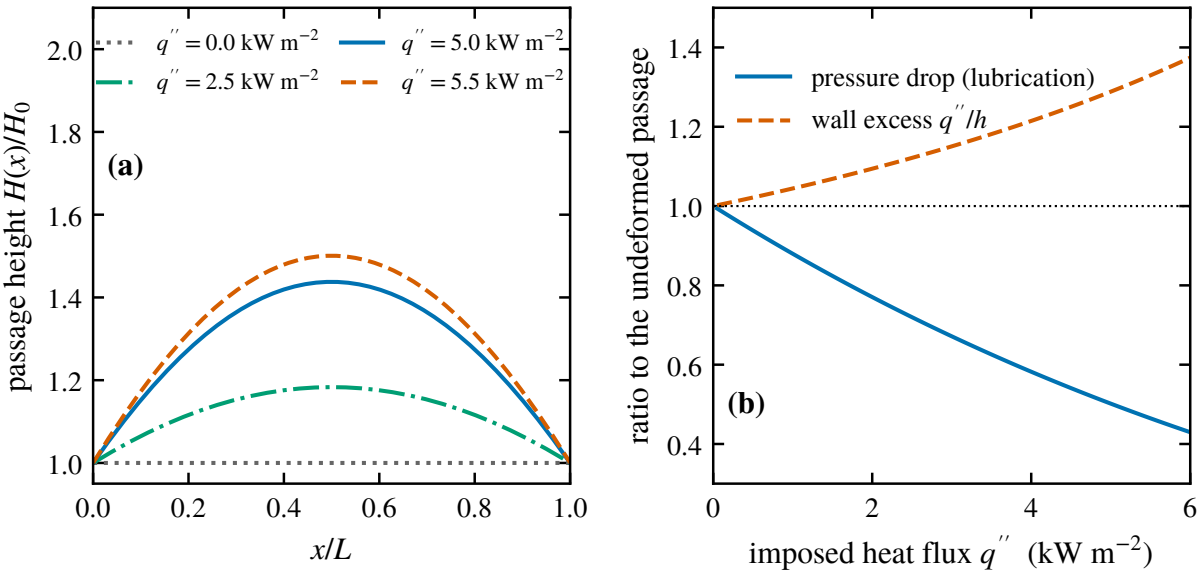



**Figure 3.** Reduced-order consequences of the deformation (indicative). (a) Passage height $H(x)/H_0$ for several imposed fluxes (frozen-force bow). (b) Lubrication estimate of the pressure drop and quasi-developed estimate of the mean interface excess $q''/h$, both relative to the undeformed passage.

## 5. Finite-element formulation and verification

This section specifies the finite-element method for the coupled problem of Section 2, states what has and has not been verified, and reports the verification of the wall idealisation. **The coupled fluid–structure–thermal solver is specified here but no result of it is reported.** The only finite-element computations in this paper are those of the wall in Section 5.4.

### *5.1 Discretisation*

*Meshes.* The fluid and the solid are meshed separately but conformingly: the two meshes share the nodes of $\Gamma_{\mathrm{if}}$, so that interface data need no interpolation. The fluid mesh follows the deformed interface through the extension problem of Section 2.3.

*Fluid.* Velocity and pressure use the Taylor–Hood pair, continuous quadratic velocity and continuous linear pressure (P2–P1), which satisfies the discrete inf–sup condition (Hood and Taylor, 1974, Brezzi and Fortin, 1991). On $\Omega_f$ the weak form seeks $(\boldsymbol{u}, p)$ with $\boldsymbol{u}$ satisfying B1 and B3–B4 and the interface condition $\boldsymbol{u} = \boldsymbol{0}$ such that, for all admissible $(\boldsymbol{v}, q)$,

$$\int_{\Omega_f} [\rho(\boldsymbol{u}\cdot\nabla\boldsymbol{u})\cdot\boldsymbol{v} + 2\mu\,\boldsymbol{D}(\boldsymbol{u}):\boldsymbol{D}(\boldsymbol{v}) - p\,\nabla\cdot\boldsymbol{v} + q\,\nabla\cdot\boldsymbol{u}]\,\mathrm{d}\Omega = 0, \tag{20}$$

where the do-nothing outflow B2 is the natural condition. The convective term is linearised by Newton iteration.

*Energy.* Temperature is one continuous quadratic field $T$ on $\Omega_f \cup \Omega_s$ with the conductivity $k = k_f$ or $k_s$ and the velocity $\boldsymbol{u}$ in the fluid and zero in the solid. Because the meshes are conforming, temperature continuity (9) is enforced strongly and the flux continuity (10) is the natural interface condition: there is a *single* interface temperature unknown and no thermal interface iteration. With B5 the weak form is

$$\int_{\Omega_f\cup\Omega_s} [\rho c_p(\boldsymbol{u}\cdot\nabla T)\,\phi + k\nabla T\cdot\nabla\phi]\,\mathrm{d}\Omega + \sum_K \int_K \tau_K\,\rho c_p\,\boldsymbol{u}\cdot\nabla\phi\,\mathscr{R}_K(T)\,\mathrm{d}\Omega = \int_{\Gamma_h} q''\,\phi\,\mathrm{d}\Gamma, \tag{21}$$

with the element residual $\mathscr{R}_K(T) = \rho c_p\,\boldsymbol{u}\cdot\nabla T - k_f\nabla^2 T$. The streamline-upwind/Petrov–Galerkin (SUPG) term (Brooks and Hughes, 1982) is required: the Péclet number is $\mathrm{Pe} = 348$, so that a Galerkin discretisa-

tion without stabilisation is expected to oscillate on any mesh whose streamwise spacing is not of micrometre order. The stabilisation parameter is $\tau_K = \frac{h_K}{2|\boldsymbol{u}|_K}\,(\coth \mathrm{Pe}_K - 1/\mathrm{Pe}_K)$ with $\mathrm{Pe}_K = \rho c_p |\boldsymbol{u}|_K h_K/(2k_f)$ and $\tau_K = 0$ in the solid.

*Solid.* The displacement is continuous quadratic. To retain the beam–column amplification of Section 3.4 the solid is solved in a total-Lagrangian description with the Green–Lagrange strain $\boldsymbol{E} = \frac{1}{2}(\boldsymbol{F}^{\mathsf{T}}\boldsymbol{F} - \boldsymbol{I})$, the deformation gradient $\boldsymbol{F} = \boldsymbol{I} + \nabla \boldsymbol{d}$, and the St Venant–Kirchhoff law with the in-plane thermal eigenstrain of A1,

$$\boldsymbol{S} = \mathbb{C} : [\boldsymbol{E} - (1+\nu)\alpha\,\theta\,\boldsymbol{I}], \qquad \int_{\Omega_s^0} \boldsymbol{S} : \delta\boldsymbol{E}\,\mathrm{d}\Omega = \int_{\Gamma_{\mathrm{if}}^0} \boldsymbol{t}\cdot\delta\boldsymbol{d}\,\mathrm{d}\Gamma, \tag{22}$$

where, for small strain and rotation, (22) reduces to the linear system (5), and $\theta = T_s - T_{\mathrm{in}}$ is the temperature rise and $\boldsymbol{t}$ is the fluid traction pulled back to the reference interface. The tangent of (22) contains a material part and a geometric part that carries the thermal membrane stress; the latter is what produces the amplification $1/(1 - N_c/N_{\mathrm{cr}})$. A linear solid omits it (Section 5.4.2).

### *5.2 Coupled fixed-point algorithm*

The mechanical coupling is a Dirichlet–Neumann partition: the fluid receives the interface displacement through the mesh extension, and the solid receives the interface traction. With the thermal problem solved as one field, the outer iteration $k = 0, 1, \ldots$ on the deformed geometry $\Omega^k$ ($\Omega^0$ undeformed) is:

1. solve the Navier–Stokes problem (20) on $\Omega_f^k$ for $(\boldsymbol{u}^k, p^k)$ and form the interface traction $\boldsymbol{t}^k$;
2. solve the energy problem (21) on $\Omega_f^k \cup \Omega_s^k$ for $T^k$, hence $\theta^k$ in the solid;
3. solve (22) with $\theta^k$ and $\boldsymbol{t}^k$ for $\boldsymbol{d}^{k+1}$;
4. relax, $\boldsymbol{d}^{k+1} \leftarrow \omega \boldsymbol{d}^{k+1} + (1-\omega)\boldsymbol{d}^k$ with $0 < \omega \le 1$ (Aitken updating is optional), and update the fluid mesh through the extension problem of Section 2.3;
5. stop when $\|\boldsymbol{d}^{k+1} - \boldsymbol{d}^k\| / \|\boldsymbol{d}^{k+1}\|$ and the relative interface heat-flux imbalance are below tolerances.

The guards of Section 3.6 are monitored during the iteration; in particular the mesh-quality guard G1 keeps the extension problem well posed. A partitioned thermal alternative, in which the interface temperature is imposed as a Dirichlet condition on the solid and the flux as a Neumann condition on the fluid, is possible on the same meshes but is not needed here. *No proof of convergence of the fixed point is offered.* The reduced-order estimates of Section 4.4 indicate a weak return coupling at the baseline, which suggests contraction, but this has not been tested.

### *5.3 Verification hierarchy*

Table V lists the verification levels needed before coupled results could be trusted and shows which have been executed. Only the first four concern the wall and the reduced-order model; all levels involving the fluid are planned.

### *5.4 Verification of the wall idealisation*

Two plane-strain computations with eight-node serendipity (Q8) elements, $3 \times 3$ Gauss quadrature and a sparse direct solver, are reported. Both are own implementations (`code/q8_strip.py` and `code/q8_nonlinear.py`); the second is cross-checked with an independent code in Section 5.4.3. The shape functions are checked for partition of unity and for their derivatives against central differences, and the patch test (the exact stress-free thermal-bending field must satisfy the discrete equations) is passed on every mesh with a maximum relative residual of $5 \times 10^{-13}$.

**Table V.** Verification hierarchy. "Done" marks computations reported in this paper.

| Level | Test | Reference | Status |
|---|---|---|---|
| V1 | Shape functions, patch test (Q8) | exact polynomial field | Done |
| V2 | Closed forms of Section 3 against an independent boundary-value solution | analytic/numerical BVP | Done |
| V3 | Free bow of the wall, end supports, mesh study | $\kappa_{\mathrm{th}} L^2/8$ | Done |
| V4 | Restrained nonlinear wall, amplification and membrane force; independent code; thin-strip limit | beam–column | Done |
| V5 | Rigid slit, pressure drop and velocity profile | $12\mu UL/H_0^2$ (Poiseuille) | Planned |
| V6 | Rigid slit, one wall heated, developed Nusselt number | Shah and London (1978) | Planned |
| V7 | Conjugate transfer with a rigid thick wall | Rahimimotlagh et al. (2024) | Planned |
| V8 | Pressure-loaded compliant channel, flow rate–pressure drop | Christov et al. (2018) | Planned |
| V9 | Mesh convergence of the coupled outputs ($w$, $\Delta p$, Nu) | Celik et al. (2008) | Planned |
| V10 | Iteration history and interface energy balance of the fixed point | internal consistency | Planned |

**Table VI.** Free-bow study: mid-span, mid-surface deflection normalised by $\kappa_{\mathrm{th}} L^2/8$. The order $p$, the extrapolated value and the grid-convergence index (GCI, safety factor 1.25) of the clamp are from the last three axial refinements, $N_x = 48, 96, 192$ (Celik et al., 2008).

| $N_y$ | elements ($N_x = 192$) | point pin ($N_x = 192$) | full-edge pin ($N_x = 192$) | clamp ($N_x = 24$) | clamp ($N_x = 192$) | $p$ | clamp, extrap. | GCI (%) |
|---|---|---|---|---|---|---|---|---|
| 2 | 384 | 1.00000 | 0.99793 | −0.0288 | −0.0195 | 1.7 | −0.0192 | 2.0 |
| 4 | 768 | 1.00000 | 0.99800 | −0.0253 | −0.0180 | 1.3 | −0.0175 | 3.8 |
| 8 | 1536 | 1.00000 | 0.99800 | −0.0238 | −0.0174 | 1.1 | −0.0168 | 4.2 |

### *5.4.1 Free bow and end supports*

With a linear through-thickness temperature of zero mean the beam theory gives the bow $w = \kappa_{\mathrm{th}} L^2/8$ at mid-span (Section 3.1). The finite-element value is sampled at the *fixed* physical point $(L/2, y = d/2)$ on every mesh, because a sampling height that changes with the through-thickness refinement is not a property of the mesh. Three end idealisations are compared: an ideal point pin ($u_y = 0$ at the single mid-surface node of each end edge), the pin of the paper realised as $u_y = 0$ on the whole end edge, and a clamp ($u_x = u_y = 0$ on the whole edge). Meshes range from $6 \times 2$ to $192 \times 8$ elements ($N_x \times N_y$, at most 1536 elements); the results are normalised by $\kappa_{\mathrm{th}} L^2/8$ and are in Table VI and Figure 4.

The ideal point pin returns 1.00000 on every mesh, as it must: the exact field lies in the element space, so this result tests the implementation and not the convergence. The pin of the paper, $u_y = 0$ on the whole edge, gives 0.99800 at $192 \times 4$ and 0.99800 at $192 \times 8$, a defect of 0.20 %. This is mesh-independent: the value changes by $3 \times 10^{-8}$ between $N_x = 96$ and 192 at $N_y = 8$ and by $4 \times 10^{-6}$ between $N_y = 4$ and 8 at $N_x = 192$. The defect is therefore a property of the end condition and not a discretisation error. The clamp bows slightly *towards* the channel. Beam theory gives zero, since a clamp forbids end rotation and a uniform-curvature strip then does not bow; the finite-element value converges slowly to a small negative number (Table VI, extrapolated $-0.0168$ at $N_y = 8$). The observed order is below two ($p = 1.1$ at $N_y = 8$), so the asymptotic range has not been established and the GCI of 4.2 % is indicative only. The relevant conclusions are that the pin idealisation is reproduced to 0.20 % and that the clamp, which is not used in

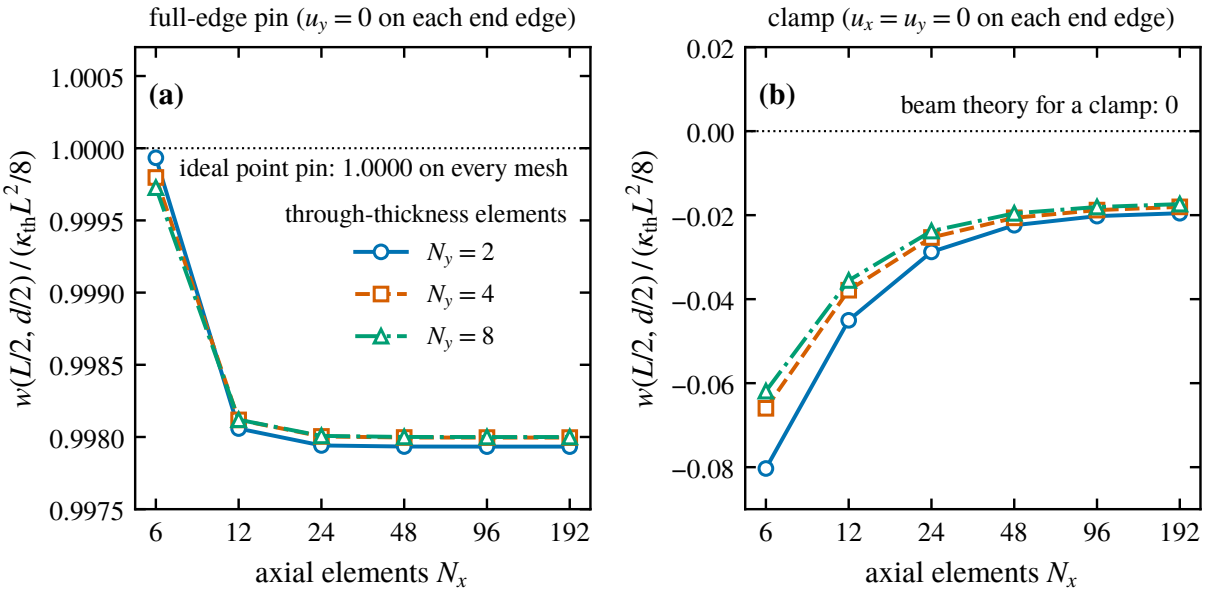



**Figure 4.** Free-bow study against axial refinement $N_x$ for $N_y = 2, 4, 8$ elements through the thickness. (a) Pin realised as $u_y = 0$ on each end edge. (b) Clamp.

**Table VII.** Restrained wall at the baseline: linear and nonlinear Q8 solutions. $w^{\mathrm{lin}}$ is the linear-solid mid-span deflection; $w^{\mathrm{NL}}_{\max}$ the total-Lagrangian one; $N_c$ is the membrane force in the two element columns adjacent to mid-span.

| mesh $N_x \times N_y$ | elements | $w^{\mathrm{lin}}/(\kappa_{\mathrm{th}} L^2/8)$ | $w^{\mathrm{NL}}_{\max}$ (µm) | $w^{\mathrm{NL}}_{\max}/H_0$ | $N_c$ at mid-span (kN m$^{-1}$) |
|---|---|---|---|---|---|
| $24 \times 2$ | 48 | 0.9979 | 223.39 | 0.4468 | 27.79 |
| $48 \times 4$ | 192 | 0.9980 | 222.46 | 0.4449 | 27.74 |
| $96 \times 4$ | 384 | 0.9980 | 222.33 | 0.4447 | 27.74 |
| $96 \times 8$ | 768 | 0.9980 | 222.17 | 0.4443 | 27.73 |
| $192 \times 8$ | 1536 | 0.9980 | 222.14 | 0.4443 | 27.73 |

this paper, reduces the bow to $-0.0174$ times the pinned bow, in agreement with beam theory (zero) to that level.

### *5.4.2 Restrained, geometrically nonlinear wall*

The baseline wall is not free: its ends are restrained axially (A7, B6), and the mean temperature rise $\bar{\theta} = 32.3$ K produces the compressive membrane force $N_c$ of Section 3.3. The second computation solves this problem: $\theta(Y) = \bar{\theta} + \Delta T_t (Y/d - \frac{1}{2})$ with the values of Table IV, $u_y = 0$ on the end edges, and a linear constraint that makes the mean axial displacement of each end edge zero, eliminated exactly. The solid is the total-Lagrangian model (22) without fluid load, solved by Newton iteration in four thermal load steps. The analytic tangent agrees with a central finite difference of the residual to $2 \times 10^{-5}$, and every step converged in at most 5 iterations to a relative residual of $4 \times 10^{-11}$. The same data are solved with the linear solid (material stiffness only) for comparison; Table VII and Figure 5 show the results.

The results are as follows. (i) The linear-solid bow is $0.9980\,\kappa_{\mathrm{th}} L^2/8$, equal to the full-edge pin value of Section 5.4.1: in a linear solid the membrane force does not couple to the bending. (ii) The linear solution carries the membrane force 29.67 kN m$^{-1}$ at mid-span, 2.2 % above the frozen-force beam value of Table IV. (iii) The nonlinear solid bows by $2.221 \times 10^{-4}$ m ($0.444\,H_0$), an amplification 1.413 over the linear value, compared with 1.39 (frozen force) and 1.36 (self-consistent) in the beam–column model.

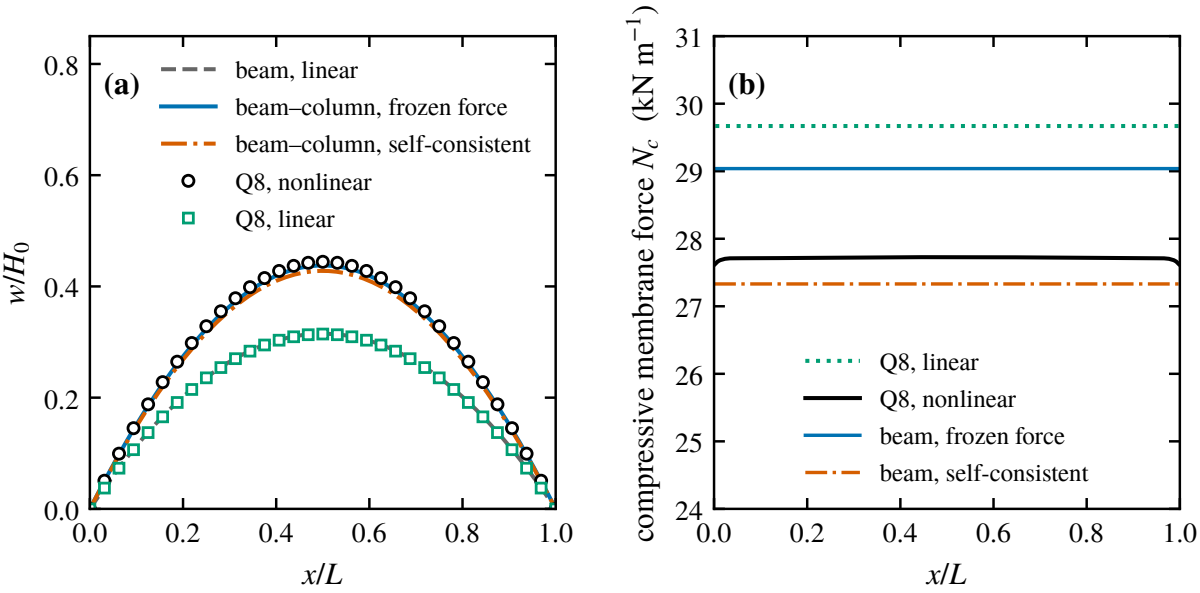



**Figure 5.** Restrained baseline wall, finest mesh ($192 \times 8$). (a) Mid-surface deflection. (b) Compressive membrane force per element column. Beam values are those of Table IV.

A linear finite-element solid therefore under-predicts the bow by a factor 1.413, which is why the total-Lagrangian form is specified in Section 5.1. (iv) The membrane force relaxes from 29.67 to 27.73 kN m$^{-1}$, that is by 6.5 %, against 5.9 % in the beam model: the arc-stretching relaxation of Section 3.4 is reproduced in magnitude. (v) The finite-element deflection exceeds the self-consistent beam–column value by 3.8 % and the frozen-force value by 1.5 %. (vi) The nonlinear result changes by 0.014 % between $N_x = 96$ and 192 and by 0.07 % between $N_y = 4$ and 8, so it is mesh-converged at this level.

### *5.4.3 Independent solution and the thin-strip limit*

*Independent code.* The same boundary-value problem was solved with a different code base and discretisation: scikit-fem, six-node quadratic triangles, Lagrange-multiplier constraints for the end conditions, a different mesh (3072 triangles at the finest level) and different quadrature (`code/fe_independent.py`). It gives $2.222 \times 10^{-4}$ m for the nonlinear bow, which differs from the Q8 value by 0.018 %, and the mid-span membrane force differs by 0.01 %; the linear solutions agree to within $10^{-4}$ %. The result of the triangle mesh itself changes by 0.03 % between the last two refinements. The two codes share the equations (St Venant–Kirchhoff kinematics, eigenstrain model), so this agreement verifies the implementation and the discretisation but not the modelling assumptions.

*Is the departure from the beam model a thickness effect?* Beam theory contains neither shear flexibility nor end effects, so if the 3.8 % of item (v) were due to the thickness of the strip ($d/L = 0.1$) it must vanish as $d/L \to 0$ at fixed dimensionless loading. The study of Table VIII keeps the free curvature and the force ratio $N_c/N_{\mathrm{cr}} = 0.275$ fixed (so $\bar{\theta} \propto d^2$ and $\Delta T_t \propto d$) and varies $d$ with the independent code. The departure from the self-consistent beam–column value falls from 3.8 % at $d/L = 0.1$ to 0.9 % at $d/L = 0.050$ and $-0.2$ % at $d/L = 0.025$, roughly as $(d/L)^2$, and the finite-element membrane force tends to the self-consistent beam force. At $d/L = 0.025$ the result changes by only 0.01 % between the $192 \times 8$ and $384 \times 16$ meshes, so the residual of $-0.2$ % is not a mesh error. It is of the size expected of the higher-order rotation terms that the von Kármán beam–column model omits (the maximum slope is 0.028), but this was not isolated. The departure of the baseline is therefore a finite-thickness effect of the two-dimensional wall, and not a disagreement between the equations of the beam model and those of

**Table VIII.** Thin-strip study with the independent code: curvature $\kappa_{\mathrm{th}} = 1.4\ \mathrm{m}^{-1}$ and $N_c/N_{\mathrm{cr}} = 0.275$ held fixed, mesh $192 \times 8$. Columns: linear bow relative to $\kappa_{\mathrm{th}} L^2/8$, nonlinear bow relative to the self-consistent (sc) and frozen-force (ff) beam–column values, and membrane force relative to the self-consistent beam force.

| $d/L$ | $d$ (mm) | $w_{\mathrm{FE}}^{\mathrm{lin}}/w_{\mathrm{beam}}^{\mathrm{lin}}$ | $w_{\mathrm{FE}}^{\mathrm{NL}}/w_{\mathrm{sc}}$ | $w_{\mathrm{FE}}^{\mathrm{NL}}/w_{\mathrm{ff}}$ | $N_{c,\mathrm{FE}}/N_c^{\mathrm{sc}}$ |
|---|---|---|---|---|---|
| 0.100 | 3.00 | 0.9980 | 1.0383 | 1.0152 | 1.0146 |
| 0.075 | 2.25 | 0.9989 | 1.0233 | 0.9846 | 1.0083 |
| 0.050 | 1.50 | 0.9995 | 1.0092 | 0.9330 | 1.0021 |
| 0.035 | 1.05 | 0.9998 | 1.0017 | 0.8720 | 0.9993 |
| 0.025 | 0.75 | 0.9999 | 0.9976 | 0.7990 | 1.0010 |

the finite-element model. The finite-element value, in contrast, lies 20 % below the frozen-force value at $d/L = 0.025$: its proximity to the baseline finite-element result (1.5 %) is a coincidence of thickness, not an accuracy property.

For the baseline strip the reduced-order deflection of Section 4 is therefore a slight under-estimate, by 3.8 %, of the two-dimensional wall response. For design use the finite-element value $0.444\,H_0$ should be preferred for this thickness; the guard values and ceiling heat fluxes derived from the reduced-order model are correspondingly slightly non-conservative.

### *5.5 What has been verified*

The reduced-order model of the wall is verified against an independent boundary-value solution (V2) and, in two dimensions, against finite elements (two independent codes) for the free bow, the end condition, the membrane force and the nonlinear amplification, including the limit of a slender strip (V1, V3, V4). The constitutive and kinematic assumptions common to all the wall models are not independently validated against experiment. The fluid-side estimates of Section 4.4, the coupling algorithm and the formulation of the interface remain unverified until V5–V10 are carried out.

## 6. Conclusions and limitations

A thin thermoelastic strip heated on its outer face bows *away from the channel*: the through-wall temperature difference sets a free curvature $\kappa_{\mathrm{th}} = (1+\nu)\alpha q''/k_s$ that is independent of the strip thickness, and pinned ends turn it into a bow that widens the passage. At the baseline the reduced-order bow is $0.428\,H_0$ and a two-dimensional geometrically nonlinear finite-element solution of the wall gives $0.444\,H_0$, which is 3.8 % higher; the beam–column amplification of the free bow (1.413 in the finite-element solution) and the relaxation of the membrane force (6.5 % against 5.9 %) are both reproduced in magnitude. The binding limits on the heat flux are deformation and stress, with buckling an outer ceiling. Indicative flow-side estimates, which need the coupled computation for confirmation, point to a hydraulic effect of order one and a notable rise of the wall excess temperature, while the effect of the flow back on the bow is small at the baseline.

The verification is partial. Only the wall has been verified (levels V1–V4 of Table V); the fluid, the interface formulation and the coupled algorithm are specified and unverified (V5–V10).

*Limitations*

- **No coupled results.** Every flow-side number is a reduced-order estimate (Section 4.4), and the lubrication and quasi-developed assumptions are not justified at the baseline parameters.
- **Material data** are representative of an unfilled PEEK-class polymer and not frozen against a datasheet; they are constant with temperature, and the strip is linearly thermoelastic (A3).
- **Idealisations.** Perfect thermal contact (A6), ideal pinned ends (A7), an idealised uniform heater (A8),

and a casing that is only a support, with no contact with the deformed strip and no clearance (A9). A real mounting would change the end conditions and, with them, the bow (Section 5.4.1).

- **Heated face.** The model heats the outer face. Heating the fluid-facing face would bow the strip into the channel and is a different model, outside this paper.
- **Wall thickness.** Plane strain is assumed. At $d/L = 0.1$ the beam–column model under-predicts the two-dimensional bow by 3.8 %; this is a finite-thickness effect that vanishes for slender strips (Table VIII).
- **Steady state.** Transient heating, thermal inertia and viscoelasticity of the polymer are not modelled.

*Further work*

Carry out V5–V10 of Table V; compute the coupled problem and test the contraction of the fixed point; replace the idealised heater and end conditions with a specified mounting and casing contact; and validate against experiment.

## A. Nomenclature

**Table IX.** Principal symbols (SI units).

| Symbol | Meaning | Symbol | Meaning |
|---|---|---|---|
| $H_0$ | undeformed passage height | $\kappa_{\mathrm{th}}$ | free thermal curvature |
| $L$, $d$ | strip length and thickness | $\nu$, $E$ | Poisson ratio, Young modulus |
| $L_e$, $L_o$ | entrance and outlet lengths | $\alpha$ | thermal expansion coefficient |
| $U$ | mean inlet velocity | $k_s$, $k_f$ | solid and fluid conductivity |
| $q''$ | imposed heat flux (supplied) | $D_s$ | flexural rigidity per unit depth |
| $\boldsymbol{u}$, $p$ | velocity, pressure | $N_c$, $N_{\mathrm{cr}}$ | compressive membrane force, Euler load |
| $T$, $\theta$ | temperature, rise above inlet | $M_b$ | buckling margin |
| $\bar{\theta}$ | mean wall temperature rise | $w_{\max}$ | maximum deflection (away from channel) |
| $\Delta T_t$, $\Delta T_b$ | through-wall and bulk temperature rise | $\Gamma_p$ | thermal-to-hydraulic deflection ratio |
| Re, Pr, Pe | Reynolds, Prandtl, Péclet numbers | Ri, Br | Richardson, Brinkman numbers |
| $\boldsymbol{d}$, $\boldsymbol{\sigma}$, $\boldsymbol{\varepsilon}$ | displacement, stress, strain | $\boldsymbol{F}$, $\boldsymbol{E}$, $\boldsymbol{S}$ | deformation gradient, Green–Lagrange strain, second Piola–Kirchhoff stress |
| $\Gamma_{\mathrm{if}}$, $\Gamma_h$ | interface, heated outer face | $\boldsymbol{n}$, $\boldsymbol{n}_{\mathrm{out}}$ | interface normal, outward solid normal |

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